

LAB 3 DELIVERABLES - COMPONENT MODELLING AND ARCHITECTURE PATTERN SELECTION

AADHAVAN MUTHUSAMY

PES1UG24CS002

PROJECT TITLE : AUTOMATED RUBRIC ASSIGNMENT EVALUATOR

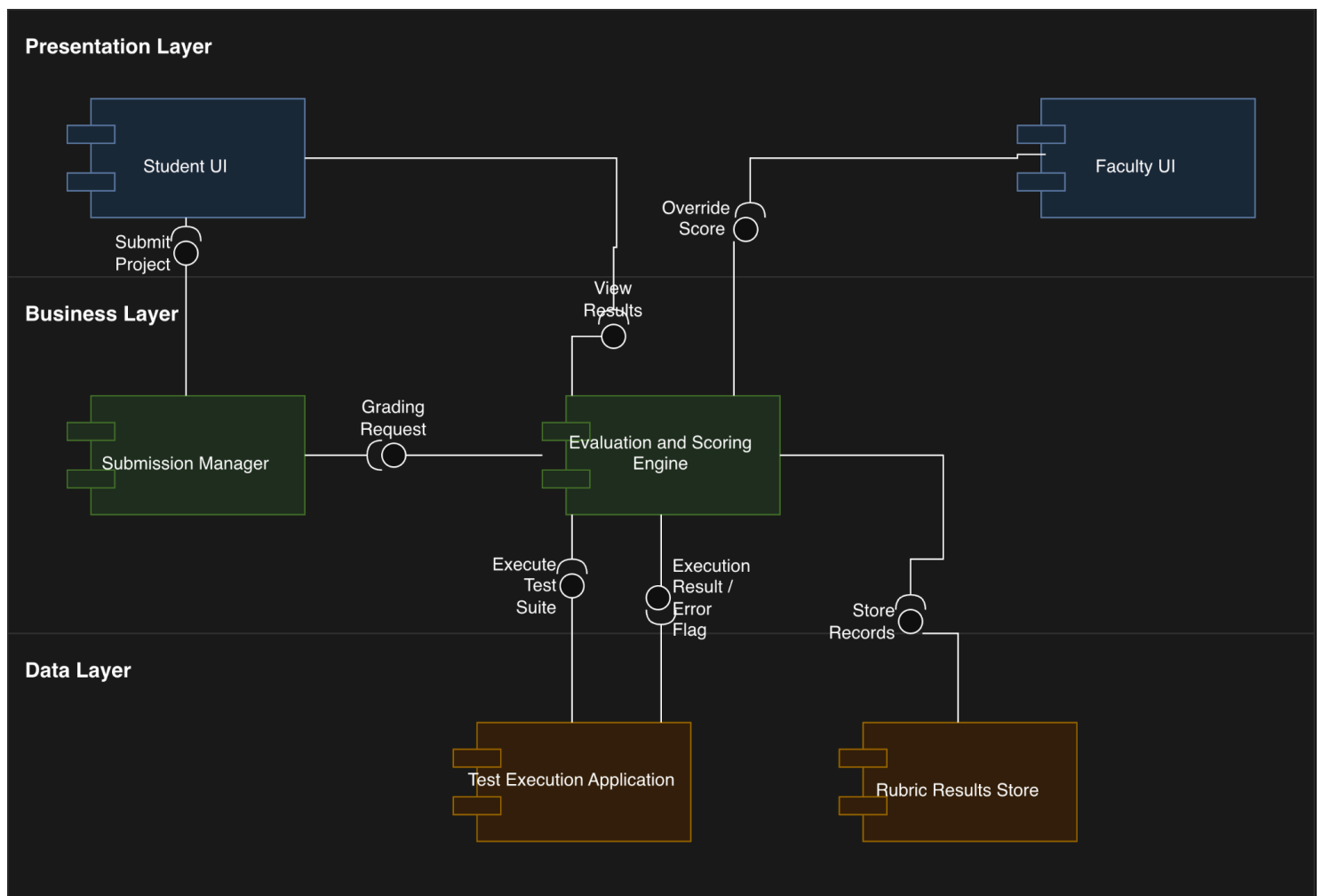
DATE : 9th SEPTEMBER 2026

SUBMITTED BY : AADHAVAN MUTHUSAMY (paya5am)

DELIVERABLES

1. ARCHITECTURE SELECTION AND DIAGRAM
2. JUSTIFICATION

ARCHITECTURE SELECTION AND DIAGRAM



JUSTIFICATION

Architecture Selection: "We chose Layered Architecture for the Automated Rubric Assignment Evaluator System."

Architectural Choice:

We selected the Layered Architecture pattern.

Two Reasons:

1. Clear Separation of Concerns: This pattern effectively isolates the user-facing interfaces (Student UI and Faculty UI) from the complex backend testing logic.
2. Supports Traditional Workflows: The step-by-step pipeline from file upload (Presentation), to queuing and grading (Business), to persistent storage (Data) maps perfectly to horizontal layers.

Security Advantage:

The architecture provides a significant security advantage by preventing direct database access from the Presentation layer. Because tight coupling is managed between specific layers, the Student UI and Faculty UI cannot directly edit the Rubric Results Store. All grade modifications must pass through the Evaluation and Scoring Engine in the Business layer, ensuring that every grade change is securely validated and audited before hitting the Data layer.

Performance Benefit:

A key performance benefit of this architecture is that the Business layer can act as a buffer. The Submission Manager can queue incoming requests from the Presentation layer before passing them to the Data layer's Test Execution Application. This prevents the heavy test-execution environment from crashing under a surge of concurrent student submissions at the deadline.